

Major Project



Explainable Machine Learning System

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Github link to the project:-

<https://github.com/prabhjotdakonda/Explainable-ML-Adult-Income>

EXPLAINABLE MACHINE LEARNING SYSTEM USING THE ADULT INCOME DATASET

**Tools Used: Python, Pandas, NumPy, Scikit-learn,
SHAP, LIME, Matplotlib**

ABSTRACT

Machine learning models are increasingly used to support decision-making in domains such as finance, healthcare, and employment. However, many machine learning models behave as black boxes, making it difficult to understand how predictions are generated. Lack of transparency can reduce trust and raise concerns regarding fairness and accountability.

The objective of this project is to build an Explainable Machine Learning System using the Adult Income Dataset and apply interpretability techniques to understand model predictions. The project involves data preprocessing, model training, evaluation, feature importance analysis, SHAP explanations, and LIME explanations. Logistic Regression and Random Forest classifiers are used to predict whether an individual's annual income exceeds \$50,000.

The results demonstrate how explainability techniques can improve transparency by identifying the most influential

features affecting predictions. Ethical considerations and fairness concerns are also discussed to highlight the importance of responsible AI development.

1. INTRODUCTION

Machine Learning models are capable of discovering complex patterns in data and making highly accurate predictions. Despite their effectiveness, many models provide little insight into how predictions are made. Such models are often referred to as black-box models.

Explainable Artificial Intelligence (XAI) aims to make machine learning systems transparent and understandable. Explainability techniques help users understand why a model made a specific prediction and which features contributed most to that prediction.

In this project, an Explainable Machine Learning System is developed using the Adult Income Dataset. The system predicts whether an individual's income is greater than \$50,000 per year while providing explanations for the model's decisions.

2. OBJECTIVES

The main objectives of this project are:

1. Train machine learning models for income prediction.
 2. Apply explainability techniques to understand model behavior.
 3. Identify the most influential features affecting predictions.
 4. Compare different machine learning models.
 5. Analyze transparency, fairness, and ethical considerations.
 6. Gain practical experience with Explainable AI techniques such as SHAP and LIME.
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3. DATASET DESCRIPTION

The Adult Income Dataset was obtained from the UCI Machine Learning Repository.

Dataset Link:

<https://archive.ics.uci.edu/ml/datasets/adult>

The dataset contains demographic, educational, and employment-related information about individuals.

The objective is to predict whether a person's annual income is:

- $\leq 50K$
- $> 50K$

Dataset Files Used

- adult.data → Training Dataset
- adult.test → Testing Dataset

Features Used

1. Age
2. Workclass
3. Final Weight (fnlwgt)
4. Education
5. Education Number
6. Marital Status
7. Occupation
8. Relationship

9. Race
10. Sex
11. Capital Gain
12. Capital Loss
13. Hours Per Week
14. Native Country

Target Variable:

Income

4. METHODOLOGY

4.1 Data Loading

The dataset was loaded using the Pandas library.

The training dataset was loaded from adult.data and the testing dataset was loaded from adult.test.

4.2 Handling Missing Values

The dataset contains missing values represented by the symbol '?'. These values were replaced with NaN values.

Missing values were found in:

- Workclass
- Occupation
- Native Country

Instead of removing rows, mode imputation was used.

Reason:

Removing rows could result in loss of useful information. Mode imputation preserves the dataset size while replacing missing values with the most frequently occurring category.

4.3 Feature and Target Separation

The dataset was divided into:

Feature Matrix (X)

- All input features

Target Variable (Y)

- Income

This separation was performed for both training and testing datasets.

4.4 Encoding Categorical Variables

Machine learning algorithms require numerical input.

Categorical features were converted into numerical values using Label Encoding.

The following columns were encoded:

- Workclass
- Education
- Marital Status
- Occupation
- Relationship
- Race

- Sex
- Native Country
- Income

Label encoding was fitted on training data and then applied to testing data to prevent data leakage.

4.5 Feature Scaling

StandardScaler was applied to normalize feature values.

Feature scaling ensures that features with larger numeric ranges do not dominate the learning process.

The scaler was fitted only on the training data and then applied to the testing data.

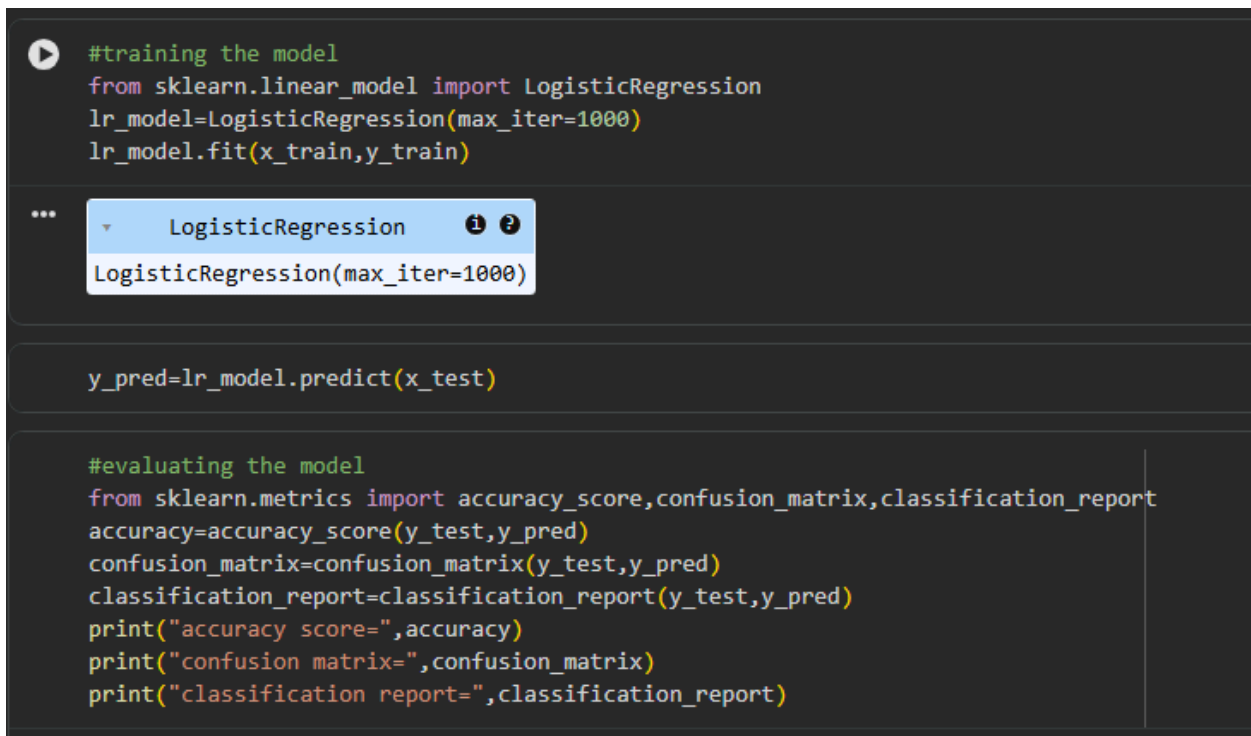
This approach prevents data leakage and ensures proper model evaluation.

5. MACHINE LEARNING MODELS

5.1 Logistic Regression

Logistic Regression was used as the baseline classification model.

The model was trained using the scaled training dataset and evaluated on the testing dataset.



```
#training the model
from sklearn.linear_model import LogisticRegression
lr_model=LogisticRegression(max_iter=1000)
lr_model.fit(x_train,y_train)

...
LogisticRegression
LogisticRegression(max_iter=1000)

y_pred=lr_model.predict(x_test)

#evaluating the model
from sklearn.metrics import accuracy_score,confusion_matrix,classification_report
accuracy=accuracy_score(y_test,y_pred)
confusion_matrix=confusion_matrix(y_test,y_pred)
classification_report=classification_report(y_test,y_pred)
print("accuracy score=",accuracy)
print("confusion matrix=",confusion_matrix)
print("classification report=",classification_report)
```

Results

Accuracy: 82.54%

Confusion Matrix:

[[11712, 723],
[2120, 1726]]

Classification Report:

Precision (Income >50K): 0.70

Recall (Income >50K): 0.45

F1-Score (Income >50K): 0.55

Interpretation

The model performed well in predicting individuals earning less than or equal to \$50,000.

However, the recall for the higher-income class was relatively low, indicating that many high-income individuals were incorrectly classified.

5.2 Random Forest Classifier

A Random Forest classifier was trained to improve predictive performance.

Random Forest combines multiple decision trees to produce more robust predictions.

Random Forest Model

```
from sklearn.ensemble import RandomForestClassifier
rf_model=RandomForestClassifier(n_estimators=100,random_state=42)
rf_model.fit(x_train,y_train)
```

RandomForestClassifier ⓘ ?

RandomForestClassifier(random_state=42)

▶ rf_pred=rf_model.predict(x_test)

```
#evaluating the random forest model
from sklearn.metrics import accuracy_score,confusion_matrix,classification_report
accuracy=accuracy_score(y_test,rf_pred)
confusion_matrix=confusion_matrix(y_test,rf_pred)
classification_report=classification_report(y_test,rf_pred)
print("accuracy score(randomforest)=",accuracy)
print("confusion matrix(randomforest)=",confusion_matrix)
print("classification report(randomforest)=",classification_report)
```

Results

Accuracy: 85.57%

Confusion Matrix:

[[11620, 815],
[1534, 2312]]

Classification Report:

Precision (Income >50K): 0.74

Recall (Income >50K): 0.60

F1-Score (Income >50K): 0.66

Interpretation

Random Forest outperformed Logistic Regression in terms of accuracy, precision, recall, and F1-score.

The model was more effective at identifying individuals with income greater than \$50,000.

6. EXPLAINABLE AI TECHNIQUES

6.1 Logistic Regression Coefficient Analysis

The coefficients of Logistic Regression were analyzed to understand feature influence.

Important positive predictors included:

- Capital Gain
- Education Number
- Age
- Sex
- Hours Per Week

These features increased the probability of predicting income greater than \$50,000.

The analysis provided an initial understanding of feature influence.

6.2 Random Forest Feature Importance

Feature importance scores were extracted from the Random Forest model.

Top features identified were:

1. fnlwgt

2. Age
3. Capital Gain
4. Relationship
5. Education Number
6. Hours Per Week

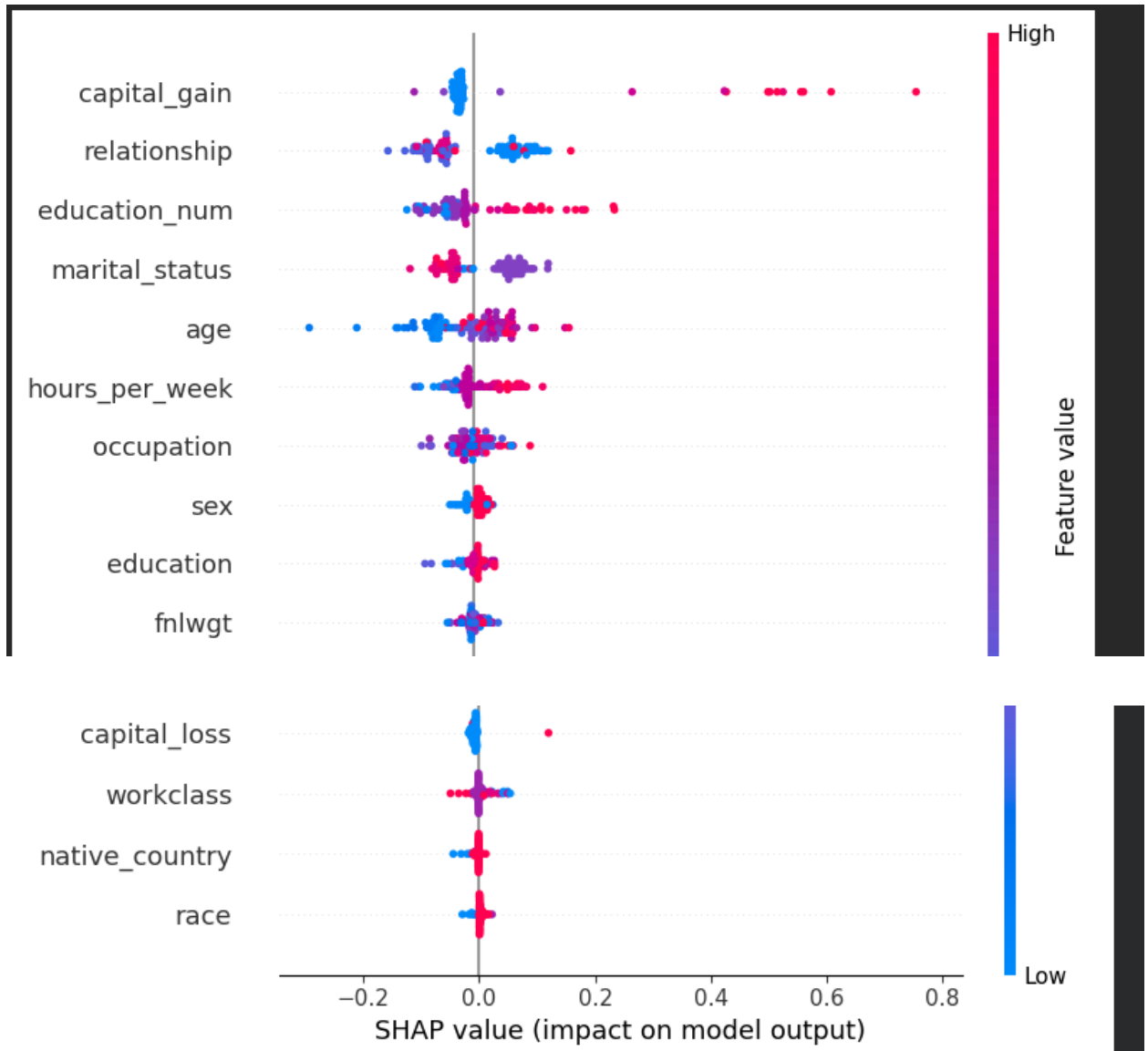
Feature importance helped identify which variables had the greatest impact on model predictions.

6.3 SHAP (SHapley Additive Explanations)

SHAP was used to provide both global and local explanations.

The SHAP summary plot revealed that:

- Capital Gain was the most influential feature.
- Education Number strongly influenced predictions.
- Age significantly affected income classification.
- Hours Per Week contributed positively to higher-income predictions.
- Relationship and Marital Status also influenced outcomes.



SHAP not only identifies important features but also shows the direction of their influence on predictions.

This improves transparency and trust in machine learning systems.

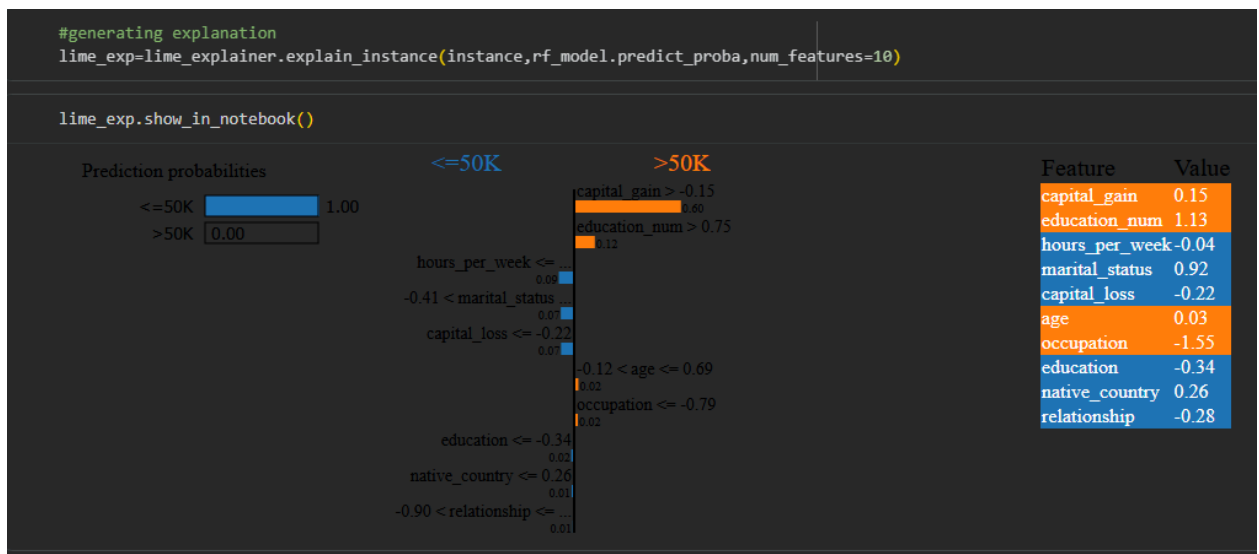
6.4 LIME (Local Interpretable Model-Agnostic Explanations)

LIME was applied to explain individual predictions.

LIME generates local explanations for specific instances by approximating model behavior around a single prediction.

Benefits of LIME:

- Explains individual predictions.
- Provides human-readable explanations.
- Improves transparency and interpretability.



LIME helped understand why a particular individual was classified as earning more than \$50,000 or less than or equal to \$50,000.

7. FAIRNESS AND ETHICAL CONSIDERATIONS

Machine learning models learn patterns from historical data.

Historical datasets may contain biases related to gender, race, education, or employment opportunities.

The analysis revealed that demographic features such as sex and race influenced predictions to some extent.

This raises important ethical concerns:

1. Fairness
2. Bias
3. Accountability
4. Transparency

Explainability techniques such as SHAP and LIME help identify whether sensitive features influence model decisions.

Responsible AI systems should be transparent and regularly audited to reduce unfair outcomes.

Machine learning predictions should support human decision-making rather than replace it entirely.

8. LEARNING OUTCOMES

Through this project, the following concepts were learned:

- Explainable Artificial Intelligence (XAI)
- Model Interpretability
- Logistic Regression
- Random Forest Classification
- Feature Importance Analysis
- SHAP Explanations
- LIME Explanations
- Data Preprocessing
- Missing Value Imputation
- Feature Scaling
- Model Evaluation
- Transparency in AI
- Fairness and Bias Analysis
- Ethical Considerations in Machine Learning

This project provided practical experience in building transparent and interpretable machine learning systems.

9. CONCLUSION

In this project, an Explainable Machine Learning System was developed using the Adult Income Dataset.

Two machine learning models, Logistic Regression and Random Forest, were trained and evaluated. Random Forest achieved the highest predictive performance with an accuracy of 85.57%.

Multiple explainability techniques, including Logistic Regression coefficient analysis, Random Forest feature importance, SHAP, and LIME, were applied to understand model behavior and feature influence.

The project demonstrated that explainability techniques can significantly improve transparency and trust in machine learning systems. Additionally, fairness and ethical considerations highlighted the importance of responsible AI development.

Overall, the project successfully achieved its objective of building a transparent and interpretable machine learning system.

REFERENCES

1. **UCI Machine Learning Repository – Adult Income Dataset**

<https://archive.ics.uci.edu/ml/datasets/adult>

2. **Scikit-learn Documentation**

<https://scikit-learn.org>

3. **SHAP Documentation**

<https://shap.readthedocs.io>

4. **LIME Documentation**

<https://lime-ml.readthedocs.io>